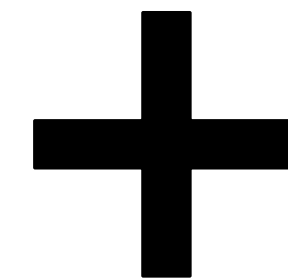
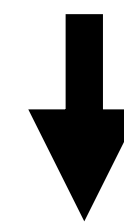


Zero-shot Prediction: Alternative to using MLM for embeddings

- (Masked) language models can also be used to directly decode the answer as a token (without fine-tuning). This is called zero-shot prediction.
- For example, this can be used to answer the question “what movies did Brad Pitt appear in?”

On November 22, 2001, Pitt made a guest appearance in the [eighth season](#) of the television series [Friends](#), playing a man with a grudge against [Rachel Green](#), played by [Jennifer Aniston](#), to whom Pitt was married at the time.^[86] For this performance he was nominated for an [Emmy Award](#) in the category of [Outstanding Guest Actor in a Comedy Series](#).^[87] In December 2001, Pitt played [Rusty Ryan](#) in the heist film [Ocean's Eleven](#), a remake of the 1960 [Rat Pack original](#). He joined an ensemble cast including [George Clooney](#), [Matt Damon](#), [Andy García](#), and Julia Roberts.^[88] Well received by critics, *Ocean's Eleven* was highly successful at the box office, earning \$450 million worldwide.^[32] Pitt appeared in two episodes of MTV's reality series [Jackass](#) in February 2002, first running through the streets of Los Angeles with several cast members in gorilla suits,^[89] and in a subsequent episode participating in his own staged abduction.^[90] In the same year, Pitt had a cameo role in George Clooney's directorial debut [Confessions of a Dangerous Mind](#).^[91] He took on his first voice-acting roles in 2003, speaking as the titular character of the [DreamWorks](#) animated film [Sinbad: Legend of the Seven Seas](#)^[92] and playing [Boomhauer](#)'s brother, [Patch](#), in an episode of the animated television series [King of the Hill](#).^[93]

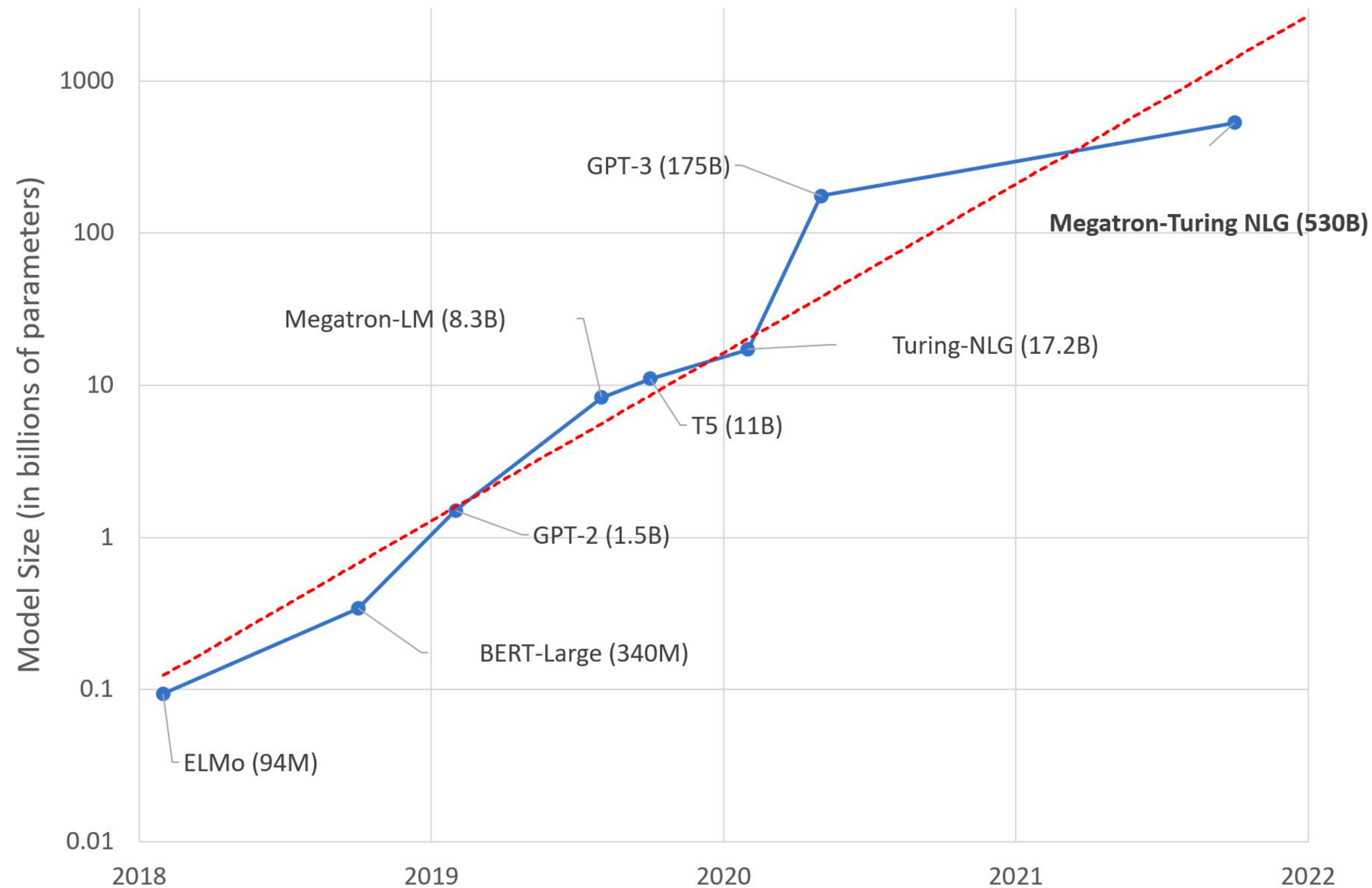
Pitt appeared in the movie [MASK]



0.3	Ocean's Eleven
0.15	Confessions of a Dangerous Mind
0.01	Friends
0.01	Jackass
...	

decoder output

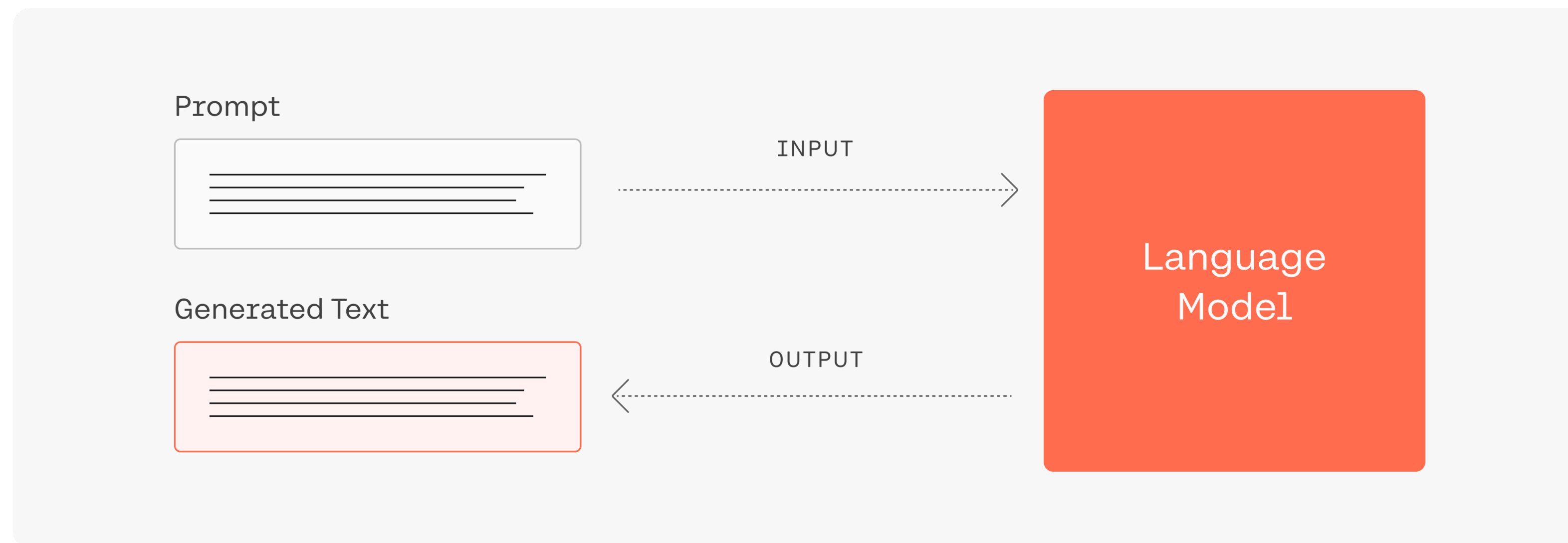
Very Large Language Models



and they are continuing to get bigger since...

Prompting

- Large language models (to give a ballpark: > 1B parameters) have become good enough at some point, that instead of training them, people began to query them directly
 - “Prompt” is the technical name of the prefix of the text given



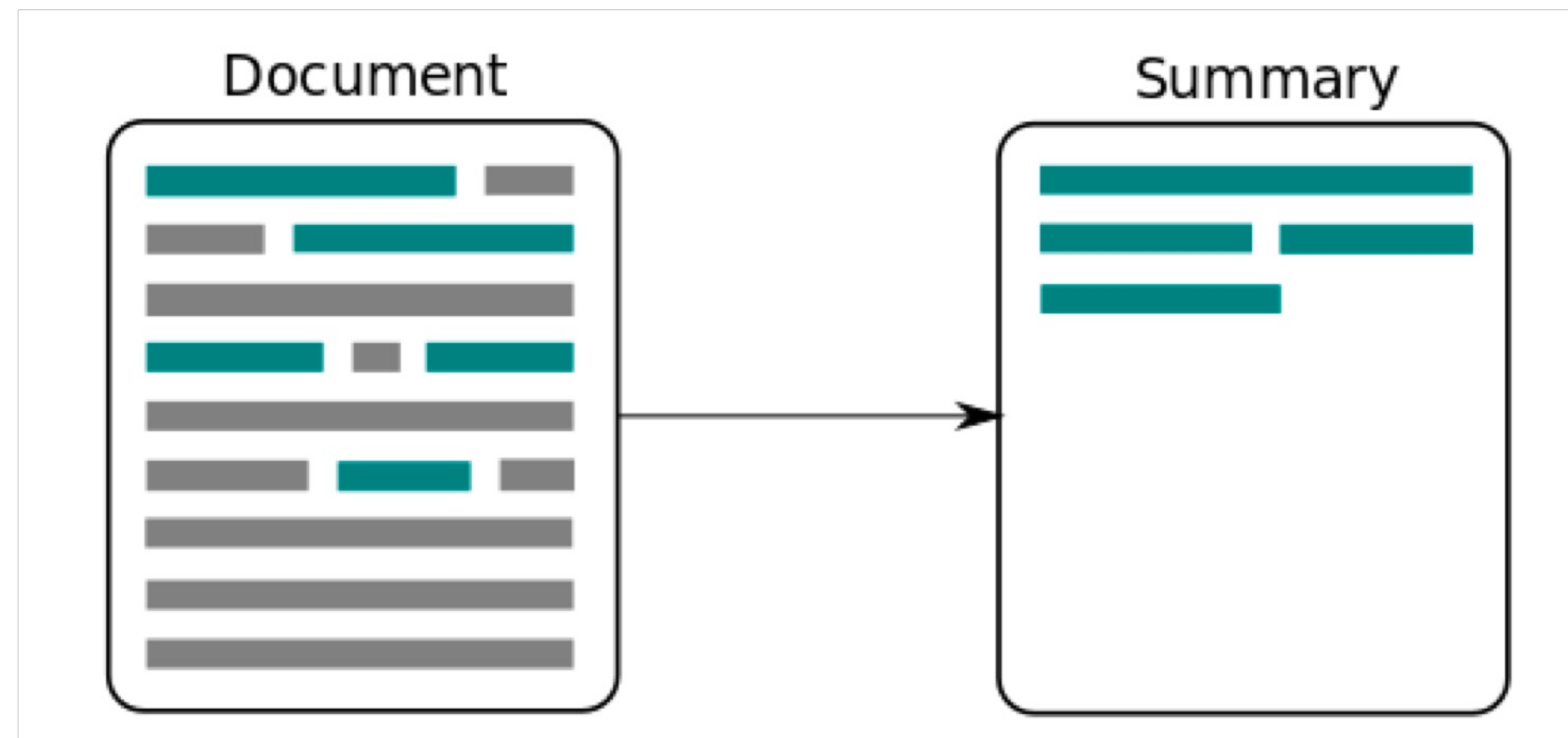
Prompting

Q: how big is France?

A:



543,940 km²



Few-shot Learning

A "whatpu" is a small, furry animal native to Tanzania. An example of a sentence that uses the word whatpu is:

We were traveling in Africa and we saw these very cute whatpus.

To do a "farduddle" means to jump up and down really fast. An example of a sentence that uses the word farduddle is:

One day when I was playing tag with my little sister, she got really excited and she started doing these crazy farduddles.

A "yalubalu" is a type of vegetable that looks like a big pumpkin. An example of a sentence that uses the word yalubalu is:

I was on a trip to Africa and I tried this yalubalu vegetable that was grown in a garden there. It was delicious.

A "Burringo" is a car with very fast acceleration. An example of a sentence that uses the word Burringo is:

In our garage we have a Burringo that my father drives to work every day.

A "Gigamuru" is a type of Japanese musical instrument. An example of a sentence that uses the word Gigamuru is:

I have a Gigamuru that my uncle gave me as a gift. I love to play it at home.

To "screeg" something is to swing a sword at it. An example of a sentence that uses the word screeg is:

We screeghed at each other for several minutes and then we went outside and ate ice cream.

Few-shot
learning: using a
new word in a
sentence

Language Models are Few-Shot Learners
Brown et al., 2020
<https://arxiv.org/pdf/2005.14165.pdf>

Zero-shot Learning

- Zero-shot capabilities: a domain-general language model (can do anything if asked to)
- More of an ideal than an accomplished goal

Context →	Please unscramble the letters into a word, and write that word: asinoc =
Target Completion →	casino

Figure G.19: Formatted dataset example for Cycled Letters

Zero-shot Learning

- Zero-shot capabilities: a domain-general language model (can do anything if asked to)
- More of an ideal than an accomplished goal

Context →	Passage: Saint Jean de Brébeuf was a French Jesuit missionary who travelled to New France in 1625. There he worked primarily with the Huron for the rest of his life, except for a few years in France from 1629 to 1633. He learned their language and culture, writing extensively about each to aid other missionaries. In 1649, Brébeuf and another missionary were captured when an Iroquois raid took over a Huron village . Together with Huron captives, the missionaries were ritually tortured and killed on March 16, 1649. Brébeuf was beatified in 1925 and among eight Jesuit missionaries canonized as saints in the Roman Catholic Church in 1930. Question: How many years did Saint Jean de Brébeuf stay in New France before he went back to France for a few years? Answer:
Target Completion →	4

Figure G.20: Formatted dataset example for DROP